



PRIYANSHU YADAV

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SUMMARY

As a PhD Research Scholar in Mathematics at BITS Pilani, my current research focuses on developing numerical solutions for fractional PDEs. I hold a strong foundational background in Lie symmetry analysis, Lie groups, and manifolds, particularly applied to nonlinear PDEs. I am passionate about bridging theoretical mathematics with practical, computational solutions, leveraging tools such as MATLAB, Mathematica, Maple, and Python to model and visualize complex systems.

WORK EXPERIENCE

PhD Research Scholar & Teaching Assistant

Jan 2026 - Present

- Serve as a Teaching Assistant at BITS Pilani and lead class discussion sessions for graduate mathematics students.
- Supervise mathematical laboratory classes, assisting students with computational problem-solving and software tools.

Research experience

Aug 2024 - Jan 2026

- Working on a research paper on Lie symmetries analysis under a professor's guidance.
- Completed and presented a master's thesis on Lie Symmetries analysis of non-linear PDE.
- Participated in a 3-day workshop on Harmonic Mappings, PDEs, and Applications at Indian Institutes of Technology, Bhubaneswar.
- Participated in a 7-day workshop on "Partial Differential Equations: Theory and Numerics" at Indian Institutes of Information Technology, Delhi

Leadership and Extracurricular Activities

Jan 2021 - May 2025

- PRESIDENT at Department of Mathematics, RLA College, University of Delhi (2022-2023).
- SECRETARY at Trikon (Society of Department of Mathematics), Netaji Subhas University of Technology, Dwarka (2023-24).
- TREASURER at Equal Opportunities Cell, RLA College, University of Delhi (2021-2022).
- Class Representative in Master of Science, Netaji Subhas University of Technology, Dwarka (2024-2025).
- Class Representative in Master of Science, Netaji Subhas University of Technology, Dwarka (2024-2025).
- Organized poster presentation competition at Netaji Subhas University of Technology, Dwarka.
- Organized the annual mathematical fest, IMPLOSION'22 of Department of Mathematics, University of Delhi.
- Organized National Workshop on "CRYPTOGRAPHY - A Recent Trend In Mathematics", "Applications of Number Theory in Real Life" etc. at Ram Lal Anand College, University of Delhi.
- Organized some events like MEMETRIX, PRESENTUS, ENIGMA in Fest of Mathematics Department of Ram Lal Anand College, University of Delhi.

Home Tutor, Account Manager**Nov 2022 - Mar 2025**

- Delivered personalized in-home & Online tutoring to students (Grades VIII–XII) in Mathematics.
- Managed 3 Leaf Infratech & Amyra traders seller accounts on ecommerce platform (amazon & flipkart) and developed and executed strategic plans for each account, analyzing and optimizing product listings, pricing strategies, and advertising campaigns. (Nov 2022 - Jan 2024)

EDUCATION

Doctor of philosophy**Jan 2026 - Present**

Birla Institute of Technology and Science, Pilani

- Research Topic: Numerical Solution of Fractional ODE/PDE.
- Integro Fractional Differential Equation.
- Modelling in Fractional Differential Equation.

Master of Science in Mathematics**Aug 2023 - May 2025**

Netaji Subhas University of Technology

CGPA: 6.5/10

- Major in Mathematics.
- Minor in Physics.
- Thesis on "Lie Symmetry analysis, Exact, Optimal solution of nmCBS equation".

Bachelor of Science Honors in Mathematics**Aug 2020 - May 2023**

University of Delhi

CGPA: 8.5/10

- Major in Mathematics.
- Minor in Computer Science.

ADDITIONAL INFORMATION

- **Languages:** English, Hindi.
- **Technical Skills:** Proficient in Matlab, MAPLE, Mathematica, LaTeX, R, Python, and C++ .
- **Short Course (40 hours):**
 - Latex: The Mathematical Typing Software.
 - Python Programming - A Multidisciplinary Approach.
 - Advance Excel and Data Analysis using Excel.
 - Full Stack Web Development and Hosting.
- **Area of Interest:**
 - Numerical Analysis.
 - Ordinary and Partial Differential Equations.
 - Linear algebra.
 - Group Theory.
 - Graph Theory.
 - Linear Programming.
 - Calculus.
 - Optimization.
 - Financial mathematics.

Webinar/Workshop

- Attended the Webinar on “Mathemagical Symposium with Prof. Brian Winkel”.
- participated in the Webinar on the topic "APPLICATIONS OF NUMBER THEORY IN REAL LIFE”
- Attended a national level webinar on "Mathematics Behind Investment Decision Making".
- Participated in the webinar on "Some Concepts From Linear Algebra by Prof. Aparna Mehra".
- Participated in the National Level workshop on "CRYPTOGRAPHY : A Recent Trand In Mathematics".
- Participated in “Scape The Scavenger Hunt Competition” conducted by Affine - The mathematics association.
- Attended a National webinar on "Introduction to Data Science"
& many more....

I hereby declare that the above-mentioned details are true and correct to the best of my knowledge.

PRIYANSHU YADAV